

1 Course outline

- signal analysis
 - types of discrete-time signals, elementary properties and operations
 - Fourier transforms for discrete-time signals
- signal processing
 - linear systems
 - filters and filter design
- from DSP to machine learning
 - random signals, noise, quantization
 - adaptive filters
- sampling and interpolation
 - from analog to digital and vice versa

2 Useful sequences

Unit impulse | Delta

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}, \quad \delta_m[n] = \delta[n - m]$$

Unit step

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Exponential decay

$$x[n] = a^n u[n], \quad 0 < a < 1$$

Sinusoidal oscillation

$$x[n] = A \cos(\omega n + \phi), \quad A \in \mathbb{R}, \omega \in \mathbb{R}, \phi \in \mathbb{R}$$

Complex exponential

$$x[n] = A e^{j(\omega n + \phi)}, \quad A \in \mathbb{R} \text{ amplitude, } \phi \text{ phase, } \omega \in \mathbb{R} \text{ frequency (rad)}.$$

Key differences

- $\delta[n]$ is the most concentrated signal in time, $u[n]$ is the most concentrated in frequency.
- $\delta[n]$ has all frequencies equally present, $u[n]$ has only DC.
- $\delta[n]$ is the identity for convolution, $u[n]$ is the identity for multiplication.

Energy and Power

- $E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$
- $P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$.
- $E_x < \infty$ and $P_x = 0 \Rightarrow$ Energy signal
- $P_x < \infty$ and $E_x = \infty \Rightarrow$ Power signal

3 Basic operations

Amplitude scaling

$$y[n] = \alpha x[n]$$

Sum of signals

$$y[n] = x_1[n] + x_2[n]$$

Product of signals

$$y[n] = x_1[n] \cdot x_2[n]$$

Time shift by $k \in \mathbb{Z}$

$$y[n] = \mathcal{S}^k x[n] = x[n + k]$$

Time reversal

$$y[n] = \mathcal{R}x[n] = \mathcal{S}^{-n} x[0] = x[-n]$$

Sum of a sequence

$$\sum_{k=m}^n z^k = \begin{cases} n - m + 1 & z = 1 \\ \frac{z^m - z^{n+1}}{1 - z} & z \neq 1 \end{cases}$$

4 Vector spaces of signals

Vector space of signals

- $\ell_2(\mathbb{Z})$: space of square-summable, infinite sequences:
$$x \in \ell_2(\mathbb{Z}) \iff \|x\|^2 = \sum_{n \in \mathbb{Z}} |x[n]|^2 < \infty$$
- $L_2([a, b])$: space of square-integrable functions over the interval $[a, b]$:
$$x \in L_2([a, b]) \iff \int_a^b |x(t)|^2 dt < \infty$$

In both cases, we require vectors to have finite energy

Inner product space

- $\|x\| = \sqrt{\langle x, x \rangle}$
- $\langle x, y \rangle = \langle y, x \rangle^*$
- $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle$
- $\langle \alpha x, y \rangle = \alpha^* \langle x, y \rangle$

Orthogonal basis coordinates

For orthog. basis $B = \{\mathbf{v}_k\}_{k=0}^{N-1}$ with inner product $\langle \cdot, \cdot \rangle$, the coord. of $\mathbf{x}$ in B are:

$$[\mathbf{x}]_B[k] = \frac{\langle \mathbf{v}_k, \mathbf{x} \rangle}{\langle \mathbf{v}_k, \mathbf{v}_k \rangle}, \quad \text{i.e. } \mathbf{x} = \sum_k [\mathbf{x}]_B[k] \mathbf{v}_k$$

DFT: basis $\{\mathbf{w}_k\}$, $\|\mathbf{w}_k\|^2 = N$, so $X[k] = \langle \mathbf{w}_k, \mathbf{x} \rangle$ (the N factor is absorbed into the synthesis $1/N$).

5 The Complex Exponential

Discrete-time complex exponential

$$w[n] = A e^{j(\omega n + \phi)}, \quad A \in \mathbb{R} \text{ amplitude, } \phi \text{ phase, } \omega \in \mathbb{R} \text{ frequency (rad)}.$$

Recursive form

$$w[n + 1] = w[n] e^{j\omega}: \text{ each step rotates by } \omega \text{ on the unit circle.}$$

Periodicity
 $\mathbf{w}$ is periodic iff $\exists N \in \mathbb{Z}^+, M \in \mathbb{Z}$ such that $\omega N = 2\pi M$:

$$\omega = \frac{M}{N} 2\pi, \quad M, N \in \mathbb{Z}$$

Only oscillations with **rational multiple of 2π** are periodic.

Aliasing

Key identity

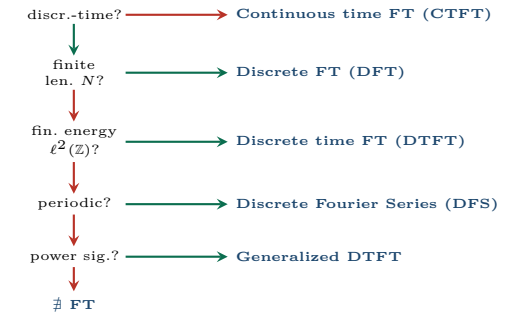
$$e^{j\theta} = e^{j(\theta + 2k\pi)} \quad \forall k \in \mathbb{Z}. \text{ Hence:}$$
$$e^{j(\omega + 2k\pi)n} = e^{j\omega n} \quad \forall k \in \mathbb{Z}.$$

Adding multiples of 2π to ω gives the **same samples**.

- $0 \leq \omega < \pi$: counterclockwise, increasing speed
- $\omega = \pi$: max speed, $+1/-1$ alternating
- $\pi < \omega < 2\pi$: more “economical” as clockwise $\omega' = \omega - 2\pi$

- Reference interval:** $[-\pi, \pi]$.
- Why complex oscillations?
- **Simpler math:** $\cos(\alpha + \beta) = \text{Re}\{e^{j\alpha} e^{j\beta}\}$
 - **Compact notation:** polar vs. two real coordinates
 - **Digital signals:** can natively be complex-valued

6 Signal analysis



7 The DFT

Fourier Basis for $\mathbb{C}^N$

Fourier basis vectors

$w_k[n] = e^{j\frac{2\pi}{N}kn}$, $n, k = 0, \dots, N-1$.
Harmonically related: $\omega_k = \frac{2\pi}{N}k$.
Note: $\mathbf{w}_k = \mathbf{w}_{N-k}^*$.

Orthogonality

$\langle \mathbf{w}_h, \mathbf{w}_k \rangle = N \delta[h-k]$, where

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n=0}^{N-1} x^*[n] y[n].$$

Basis is orthogonal but **not** orthonormal (norms = $\sqrt{N}$).

Analysis & Synthesis

DFT (analysis)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk}, \quad k = 0, \dots, N-1$$

Inverse DFT (synthesis)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk}, \quad n = 0, \dots, N-1$$

Elementary DFT Pairs

Impulse & constant (duality)

$\delta \xleftrightarrow{\text{DFT}} \mathbf{1}$ (most concentrated in time $\Rightarrow$ flat spectrum)
 $\mathbf{1} \xleftrightarrow{\text{DFT}} N\delta$ (most spread in time $\Rightarrow$ single freq.)

Shifted impulse

$$\delta[n-m] \xleftrightarrow{\text{DFT}} e^{-j\frac{2\pi}{N}mk} = \mathbf{w}_m^*[k]$$

Harmonic sinusoid

$\mathbf{w}_m \xleftrightarrow{\text{DFT}} N\delta_m$, where $w_m[n] = e^{j\frac{2\pi}{N}mn}$
Spread in time $\Rightarrow$ single nonzero freq. coefficient.

Rectangular (M -length rect.)

$x[n] = 1$ for $0 \leq n < M$, else 0:

$$X[k] = \frac{\sin(\pi M k / N)}{\sin(\pi k / N)} e^{j\frac{\pi}{N}(M-1)k}$$

Zeros at $k = mN/M$ ($m \neq 0$). $M = N$: gives $N\delta$.

Time-frequency concentration duality

Most concentrated in time $\Leftrightarrow$ most spread in frequency (and vice versa). Generalizes $\delta \leftrightarrow \mathbf{1}$ and $\mathbf{1} \leftrightarrow N\delta$.

Roots of unity

The N solutions of $z^N = 1$: $z_k = e^{j\frac{2\pi}{N}k}$, $k = 0, \dots, N-1$. These are the frequencies $\omega_k = \frac{2\pi}{N}k$ satisfying $e^{j\omega_k N} = 1$.

Matrix Form

DFT matrix

$\mathbf{X} = \mathbf{W}\mathbf{x}$, $\mathbf{x} = \frac{1}{N}\mathbf{W}^H\mathbf{X}$, where $W_N = e^{-j\frac{2\pi}{N}}$:

$$\mathbf{W} = \begin{bmatrix} W_N^0 & W_N^0 & \dots & W_N^0 \\ W_N^0 & W_N^1 & \dots & W_N^{N-1} \\ \vdots & & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & \dots & W_N^{(N-1)^2} \end{bmatrix}$$

$$\mathbf{W}^H \mathbf{W} = N\mathbf{I}.$$

Powers modulo N

$W_N^n = W_N^{n \bmod N}$ (aliasing property). First row and column are all 1 since $W_N^0 = 1$.

Small DFT matrices

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\mathbf{W}_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3 & W_3^2 \\ 1 & W_3^2 & W_3 \end{bmatrix}, \quad W_3 = \frac{-1-j\sqrt{3}}{2}$$

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

8 Properties

DFT Properties

Linearity

$$\text{DFT}\{\mathbf{a}\mathbf{x} + \mathbf{b}\mathbf{y}\} = a \text{DFT}\{\mathbf{x}\} + b \text{DFT}\{\mathbf{y}\}, \quad a, b \in \mathbb{C}$$

Signal decomposition (computation trick)

Decompose $\mathbf{x} = \sum_i a_i \mathbf{s}_i$ into signals with known DFTs, then use linearity.
Ex.: $\mathbf{x} = a\delta + b \cdot \mathbf{1} \Rightarrow \mathbf{X} = a \cdot \mathbf{1} + bN\delta$

Circular time shift

$$x[(n-n_0) \bmod N] \xleftrightarrow{\text{DFT}} e^{-j\frac{2\pi}{N}n_0k} X[k]$$

Frequency shift

$$x[n] e^{j\frac{2\pi}{N}k_0n} \xleftrightarrow{\text{DFT}} X[k-k_0]$$

Modulation by cosine

$$x[n] \cos\left(\frac{2\pi}{N}k_0n\right) \xleftrightarrow{\text{DFT}} \frac{X[k-k_0] + X[k+k_0]}{2}$$

Time/frequency reversal

$$\begin{aligned} \text{DFT}\{\mathcal{R}\mathbf{x}\} &= \mathcal{R}\mathbf{X} \\ \text{DFT}\{\mathbf{x}^*\} &= \mathcal{R}\mathbf{X}^* \end{aligned}$$

Parseval's theorem (DFT)

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 = \|\mathbf{x}\|_2^2$$

Energy is preserved across the DFT change of basis ($1/N$ from non-normalized basis). $|X[k]|^2/N$ = energy at frequency ω_k . Useful: $\|\mathbf{x} - \mathbf{y}\|_2^2 = \frac{1}{N}\|\mathbf{X} - \mathbf{Y}\|_2^2$.

Symmetries of the DFT

A signal $\mathbf{x}$ is **symmetric** if $x[n] = x[N-n]$.
A signal is **Hermitian-symmetric** if $x[n] = x^*[N-n]$.

Symmetry summary

Time domain $x[n]$	Freq. domain $X[k]$
Real	$X[-k] = X^*[k]$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

9 Reading & Plotting the DFT

DFT magnitude plot

Index k encodes frequency $\omega_k = \frac{2\pi}{N}k$:

- $k = 0$: DC (zero frequency)
- $k = 1 \dots \lfloor N/2 \rfloor$: positive (CCW) frequencies, increasing
- $k = \lfloor N/2 \rfloor$: maximum $\omega = \pi$ (fastest oscillation)
- $k = \lfloor N/2 \rfloor + 1 \dots N-1$: negative (CW) frequencies, decreasing

For real $\mathbf{x}$: often show only $k = 0 \dots N/2$ (symmetric).

DFT phase

Phase is 2π -periodic, plotted over $[-\pi, \pi]$. Values outside are **wrapped**. Phase offset θ in time shifts all components; same magnitude DFT can give different time-domain shapes.

Frequency axis in Hz

If signal sampled at $F_s = N/T$ Hz (T = duration in seconds):

$$k \rightarrow \frac{F_s}{N}k \text{ Hz}, \quad 0 \leq k \leq N/2$$

$$f_{\max} = \frac{F_s}{2} \text{ (Nyquist)}$$

For $k > N/2$: $k \rightarrow \frac{F_s}{2} - \frac{F_s}{N}k$ Hz (negative freqs).

10 DFS & DFT Periodicity

DFT implies N-periodicity

The inverse DFT formula is valid for **all** $n \in \mathbb{Z}$:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk} = x[n \bmod N]$$

Both x and X are implicitly N -periodic.

DFS (Discrete Fourier Series)

For an N -periodic signal $\tilde{x}$:

$$\tilde{\mathbf{X}} = \text{DFS}\{\tilde{\mathbf{x}}\} = \text{DFT}\{\tilde{x}[0], \dots, \tilde{x}[N-1]\}$$

DFS $\equiv$ DFT: same formulas, different interpretation (explicit periodicity).

Circular shift (DFS)

$$\tilde{x}[n-n_0] \xleftrightarrow{\text{DFS}} e^{-j\frac{2\pi}{N}n_0k} \tilde{X}[k]$$

Time shifts are circular
With finite-length signals, time shifts wrap around modulo N . A linear phase multiplication on DFT coefficients computes the circular shift of the signal.

11 The DTFT

DTFT (analysis)
For infinite-length sequence $\mathbf{x}$:
$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
$X(\omega)$ is a continuous, 2π-periodic function of $\omega \in \mathbb{R}$.

Inverse DTFT (synthesis)
$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$
Uses $\int_{-\pi}^{\pi} e^{j\omega(n-k)} d\omega = 2\pi \delta[n-k]$.

Existence & Convergence

Existence: absolutely summable $\Rightarrow$ DTFT
If $\sum_{n=-\infty}^{\infty} x[n] < \infty$, then $X(\omega)$ exists, is continuous, and $ X_N(\omega) \rightarrow X(\omega)$ uniformly.

Square-summable (finite energy)
If $\mathbf{x} \in \ell_2(\mathbb{Z})$: $\sum x[n] ^2 < \infty$, DTFT exists in mean-square sense:
$\lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} X_N(\omega) - X(\omega) ^2 d\omega = 0$
$X(\omega)$ may not be continuous (Gibbs phenomenon possible).

DTFT as Limit of DFT
As $N \rightarrow \infty$: $\omega_N = 2\pi/N \rightarrow 0$, DFT samples become dense over $[0, 2\pi]$. By substituting $\omega = 2\pi k/N$:

$$X[k] \rightarrow X\left(\frac{2\pi k}{N}\right)$$

Also: $\tilde{X}_N[k] = X\left(\frac{2\pi}{N}k\right)$ (DFS as samples of DTFT).

12 DTFT Pairs & Properties
Elementary DTFT Pairs
Impulse
DTFT $\{\delta_m\} = e^{-j\omega m}$ Most concentrated in time $\Rightarrow$ spread uniformly in freq.

Rectangular signal
$x[n] = \begin{cases} 1 & 0 \leq n < M \\ 0 & \text{otherwise} \end{cases}$
$X(\omega) = \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j\omega(M-1)/2}$
Zeros at $\omega = 2\pi m/M$, $m \neq 0$.

Exponential decay ($ a < 1$)
$x[n] = a^n u[n]$ ($u[n]$: unit step)
$X(\omega) = \frac{1}{1 - a e^{-j\omega}}$
$ X(\omega) ^2 = \frac{1}{1 + a ^2 - 2a \cos \omega}$

DTFT Properties
Linearity
DTFT $\{ax + by\} = a\mathbf{X} + b\mathbf{Y}$, $a, b \in \mathbb{C}$
Time shift
$x[n - n_0] \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(\omega)$

Frequency shift (modulation)
$x[n] e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} X(\omega - \omega_0)$
$x[n] \cos(\omega_0 n) \xleftrightarrow{\text{DTFT}} \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$

Time & frequency reversal
DTFT $\{\mathcal{R}\mathbf{x}\} = \mathcal{R}\mathbf{X}$ DTFT $\{\mathbf{x}^*\} = \mathcal{R}\mathbf{X}^*$

Symmetries of the DTFT
A signal is **even** if $x[n] = x[-n]$, **odd** if $x[n] = -x[-n]$.
A signal is **Hermitian-symmetric** if $x[n] = x^*[-n]$.

Symmetry summary	
Time domain $x[n]$	Freq. domain $X(\omega)$
Real	$X(-\omega) = X^*(\omega)$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

Parseval's theorem (DTFT)
$\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) ^2 d\omega$

Reading DTFT plots
DTFT frequency axis
Plotted over $[-\pi, \pi]$: $\omega = 0$: DC $\omega \in (0, \pi]$: positive (CCW) = high freqs near $\pm\pi$ $\omega \in [-\pi, 0)$: negative (CW) = high freqs near $\pm\pi$ - Low freqs near 0, high freqs near $\pm\pi$ DTFT is 2π -periodic; plots outside $[-\pi, \pi]$ repeat.

Generalized DTFT (Power Signals)
Via functionals for infinite-energy signals:

Constant & complex exponential
DTFT $\{1\} = \delta(\omega)$ (Dirac delta at $\omega = 0$) $e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} \delta(\omega - \omega_0)$

Sinusoids
$\cos(\omega_0 n) \xleftrightarrow{\text{DTFT}} \frac{1}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$ $\sin(\omega_0 n) \xleftrightarrow{\text{DTFT}} \frac{-j}{2} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

Unit step
$U(\omega) = \frac{1}{1 - e^{-j\omega}} + \frac{1}{2} \delta(\omega)$

13 DFT vs. DTFT vs. DFS		
	Input	Output
DFT	finite, length N	vector $\in \mathbb{C}^N$
DFS	N -periodic	N -periodic
DTFT	infinite (ℓ_2)	2π -periodic function

DFT/DFS equivalence
DFS of an N -periodic signal = DFT of one period. Both use identical formulas (4.10) and (4.11); only the interpretation of periodicity differs.

DTFT as DFS limit
$\tilde{x}_N[n] = \sum_{p=-\infty}^{\infty} x[n + pN]$ (periodization). $\tilde{X}_N[k] = X\left(\frac{2\pi}{N}k\right)$: DFS samples the DTFT. As $N \rightarrow \infty$: $\tilde{x}_N \rightarrow \mathbf{x}$ and samples fill $[0, 2\pi]$.

14 The FFT
Complexity comparison
- Naive DFT: $O(N^2)$ multiplications - FFT: $O(N \log N)$ operations - For $N = 10,000$: $\sim 40,000$ vs. > 100 million ops

Decimation in time (Cooley-Tukey)
Split $N = 2^L$ input into even/odd: $x_e[n] = x[2n]$, $x_o[n] = x[2n+1]$, $n = 0, \dots, \frac{N}{2} - 1$ Output halves: $X_a[k] = X[k]$ and $X_b[k] = X[k + \frac{N}{2}]$, $k = 0, \dots, \frac{N}{2} - 1$ Butterfly equations:
$X_a[k] = X_e[k] + e^{j\frac{2\pi}{N}k} X_o[k]$
$X_b[k] = X_e[k] - e^{j\frac{2\pi}{N}k} X_o[k]$
where X_e, X_o are the $(N/2)$ -point DFTs of x_e, x_o .

Cost breakdown
$2(N/2)^2 = N^2/2$ mults for $\mathbf{X}_e$ and $\mathbf{X}_o$ $N/2$ mults by twiddle factors for $\mathbf{X}_o$ - Second half is free (just sign change) Recursion gives $O(N \log_2 N)$ total.

15 Spectrograms (STFT)
Short-Time Fourier Transform
Analyse a signal $\mathbf{x}$ of length L time-locally:
1. Slide a window of size W to get $\lfloor L/W \rfloor$ non-overlapping bins
2. Compute a W -point DFT in each bin
3. Result: 2D time $\times$ frequency representation

Kronecker delta in spectrogram
DFT of a length- W Kronecker delta $\delta[n-m]$ (even shifted): $ X[k] = 1$ for all k . $\Rightarrow$ In a spectrogram with window W : the bin containing m has constant magnitude W across all frequencies; all other bins are 0 .

Window size trade-off
Large W : good frequency resolution, poor time resolution.
Small W : good time resolution, poor frequency resolution.
This is an instance of the time–frequency concentration duality.

16 Discrete-Time Filters

LTI Systems

Linearity & time-invariance

Let $y[n] = \mathcal{H}\{x[n]\}$.

Linearity ($\forall \alpha, \beta \in \mathbb{C}, \forall x_1, x_2$):

$$\mathcal{H}\{\alpha x_1[n] + \beta x_2[n]\} \stackrel{?}{=} \alpha \mathcal{H}\{x_1[n]\} + \beta \mathcal{H}\{x_2[n]\}$$

Time-Invariance ($\forall n_0 \in \mathbb{Z}, \forall x$):

$$\underbrace{\mathcal{H}\{x[n - n_0]\}}_{x[\cdot] \leftarrow x[\cdot - n_0]} \stackrel{?}{=} \underbrace{y[n - n_0]}_{n \leftarrow n - n_0}$$

Key consequence: an LTI system cannot create new frequency content.

Zero initial conditions

A recursive filter is LTI and has a *unique* output only if $\exists n_0$ such that for $n < n_0$: input is zero, output is zero, and all memory cells are zero. Nonzero initial memory breaks both uniqueness and time-invariance.

Impulse Response & Convolution

Impulse response

$\mathbf{h} = \mathcal{H}\delta$. An LTI system is *fully* described by $\mathbf{h}$.

Convolution $\mathbf{y} = \mathbf{x} * \mathbf{h}$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Well-defined when one sequence is abs. summable and the other is bounded (or both abs. summable).

To compute $y[n]$ sample by sample: time-reverse h , slide it to position n , multiply element-wise with x , sum.

Convolution properties

Commutativity: $\mathbf{x} * \mathbf{h} = \mathbf{h} * \mathbf{x}$

Linearity: $\mathbf{x} * (\alpha \mathbf{y} + \beta \mathbf{w}) = \alpha (\mathbf{x} * \mathbf{y}) + \beta (\mathbf{x} * \mathbf{w})$

Time-shift: $\mathbf{x} * (\mathcal{S}^k \mathbf{y}) = \mathcal{S}^k (\mathbf{x} * \mathbf{y})$

Associativity: $(\mathbf{x} * \mathbf{h}) * \mathbf{w} = \mathbf{x} * (\mathbf{h} * \mathbf{w})$ (*abs. summable*)

Cascade: filters in series $\equiv$ single filter with $\mathbf{h} = \mathbf{h}_1 * \mathbf{h}_2$; order is irrelevant.

Filter Properties (from h)

FIR vs IIR

- **FIR** (Finite Impulse Response): $\mathbf{h}$ has finite support. No feedback; *always* BIBO stable.
- **IIR** (Infinite Impulse Response): $\mathbf{h}$ has infinite support. Realizable IIRs use feedback and have a one-sided $\mathbf{h}$.

Causality

Output at time n depends only on $x[n], x[n - 1], \dots$ (past/present). Equivalently, $h[n] = 0$ for $n < n_0$ with some $n_0 \geq 0$.

- **Causal up to a delay:** $h[n] = 0$ for $n < n_0 < 0$ — FIR filters always satisfy this.
- **Anticausal:** $h[n] = 0$ for $n > 0$; offline use only.
- **Ideal:** h has infinite two-sided support — *not realizable*, used as theoretical benchmark.

BIBO Stability

LTI is BIBO stable $\iff \mathbf{h}$ is absolutely summable:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

($\Leftarrow$) If $|x[n]| \leq L$ then $|y[n]| \leq L \sum_k |h[k]| < \infty$.
($\Rightarrow$) If $\sum |h[k]| = \infty$, take $x[n] = \text{sgn}(h[-n])$; then $y[0] = \infty$.

FIR is always BIBO stable (finite sum is always finite).

Realizable Filters: CCDE

Constant-Coefficient Difference Equation

$$y[n] = \underbrace{\sum_{k=0}^{M-1} b_k x[n - k]}_{\text{feedforward (FIR part)}} - \underbrace{\sum_{k=1}^{N-1} a_k y[n - k]}_{\text{feedback (IIR part)}}$$

FIR: all $a_k = 0$ for $k \geq 1$.

IIR: at least one $a_k \neq 0$ for $k \geq 1$ — recursion introduces a feedback loop.

Frequency Response

Eigensequence property

Complex exponentials are eigenfunctions of any LTI system:

$$\mathcal{H}\{e^{j\omega_0 n}\} = H(\omega_0) \cdot e^{j\omega_0 n}$$

$$H(\omega) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$$

$|H(\omega_0)|$ scales the amplitude; $\angle H(\omega_0)$ shifts the phase. $H(\omega)$ is the DTFT of $\mathbf{h}$.

Convolution theorem

$$y = x * h \iff Y(\omega) = X(\omega) \cdot H(\omega)$$

Filtering in time = pointwise multiplication in frequency. $H(\omega)$ acts as a *spectral mask*.

Filter types by magnitude response

- **Lowpass:** passes low freqs ($\omega \approx 0$), blocks high ($\omega \approx \pm\pi$)
 - **Highpass:** passes high freqs, blocks low
 - **Bandpass:** passes freqs near $\pm\omega_c$
 - **Allpass:** $|H(\omega)| = \text{const}$ — modifies phase only
- Passband:** $|H| \approx 1$; **Stopband:** $|H| \approx 0$.

Linear phase & group delay

Linear phase: $\angle H(\omega) = -\omega d$ — all frequencies delayed by exactly d samples. Generalized: $H(\omega) = A(\omega)e^{-j(\omega d + \phi)}$, $A(\omega) \in \mathbb{R}$.

Group delay: $\text{grd}\{H(\omega)\} = -\frac{d \angle H(\omega)}{d\omega}$

Constant group delay $\iff$ linear phase. All realizable linear-phase filters are FIR.

Running Examples

Moving Average (MA)

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k], \quad h[n] = \begin{cases} \frac{1}{M} & 0 \leq n < M \\ 0 & \text{else} \end{cases}$$

Causal FIR $\Rightarrow$ always BIBO stable. Cost: $M - 1$ additions + 1 mult per sample.

Zero-centered (noncausal): $y[n] = \frac{1}{2L+1} \sum_{k=-L}^L x[n - k]$ — no lag; causal up to delay L .

MA frequency response

$$H(\omega) = \frac{1}{M} \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j \frac{M-1}{2} \omega}$$

Lowpass; $M - 1$ nulls at $\omega = 2\pi m/M$ ($m \neq 0$). Generalized linear phase, delay $\frac{M-1}{2}$. Larger $M \Rightarrow$ narrower passband, stronger smoothing.

Leaky Integrator (LI)

$$y[n] = \lambda y[n - 1] + (1 - \lambda) x[n], \quad |\lambda| < 1$$

$$h[n] = (1 - \lambda) \lambda^n u[n] \quad (\text{causal, exponential decay})$$

Causal IIR. Constant cost: 2 mult + 1 add per sample regardless of smoothing power.

Approximates MA of length M with $\lambda = \frac{M-1}{M}$.

LI frequency response & stability

$$H(\omega) = \frac{1 - \lambda}{1 - \lambda e^{-j\omega}}, \quad |H(\omega)|^2 = \frac{(1 - \lambda)^2}{1 + \lambda^2 - 2\lambda \cos \omega}$$

Lowpass; bandwidth $\downarrow$ as $\lambda \rightarrow 1$. BIBO stable $\iff |\lambda| < 1$:

$$\sum_{n \geq 0} |h[n]| = \frac{|1 - \lambda|}{1 - |\lambda|} < \infty \iff |\lambda| < 1$$

At $\lambda = 1$: perfect integrator — output diverges.

Ideal Filters

sinc & rect

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}, \quad \text{sinc}(0) = 1. \quad \text{rect}(x) = \begin{cases} 1 & |x| \leq \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

$$\text{DTFT pair: } \frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right) \xleftrightarrow{\text{DTFT}} \text{rect}\left(\frac{\omega}{2\omega_p}\right)$$

Fundamental sinc/rect DTFT pair

$$\frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right) \xleftrightarrow{\text{DTFT}} \text{rect}\left(\frac{\omega}{2\omega_p}\right)$$

As $\omega_p \rightarrow \pi$: recovers $\delta \leftrightarrow \mathbf{1}$. As $\omega_p \rightarrow 0$: $\mathbf{1} \leftrightarrow \tilde{\delta}(\omega)$.

Ideal Lowpass (cutoff ω_p)

$$H_L(\omega) = \begin{cases} 1 & |\omega| \leq \omega_p \\ 0 & \text{else} \end{cases}, \quad h_L[n] = \frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right)$$

Infinite two-sided support $\Rightarrow$ **not realizable**.

Ideal Highpass & Bandpass

$$H_H(\omega) = 1 - H_L(\omega), \quad h_H[n] = \delta[n] - h_L[n]$$

$$h_B[n] = 2 \cos(\omega_c n) \frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right) \quad (\text{BW } 2\omega_p, \text{ center } \omega_c)$$

Hilbert Filter (allpass)

$$H_F(\omega) = \begin{cases} -j & 0 \leq \omega < \pi \\ +j & -\pi \leq \omega < 0 \end{cases}, \quad |H_F| = 1$$

$$h_F[n] = \begin{cases} 2/(n\pi) & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

17 Z-Transform & Filter Design

Z-Transform

Z-Transform

$$\mathcal{Z}\{\mathbf{x}\}(z) = X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad z \in \mathbb{C}$$

A formal power series in z^{-1} ; existence requires the sum to converge absolutely for a given z .

Fundamental properties

Linearity: $\mathcal{Z}\{\alpha \mathbf{x} + \beta \mathbf{y}\} = \alpha X(z) + \beta Y(z)$
Time-shift: $\mathcal{Z}\{x[n - N]\} = z^{-N} X(z)$ (delay by N multiplies by z^{-N})
Convolution: $\mathcal{Z}\{\mathbf{h} * \mathbf{x}\} = H(z) \cdot X(z)$

Link to DTFT

Evaluating on the unit circle $z = e^{j\omega}$ recovers the DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \text{DTFT}\{\mathbf{x}\}$$

The DTFT is the z-transform restricted to $|z| = 1$ (when the unit circle is inside the ROC).

Transfer Function

Transfer function $H(z)$

The z-transform of the impulse response $\mathbf{h}$:

$$H(z) = \mathcal{Z}\{\mathbf{h}\} = \frac{Y(z)}{X(z)}$$

For a CCDE (apply z-transform to both sides):

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{B(z)}{A(z)}$$

$B(z)$: feedforward (FIR part). $A(z)$: feedback (IIR part). FIR: $A(z) = 1$.

Zeros & poles — factored form

$$H(z) = b_0 \frac{\prod_{k=0}^{M-1} (1 - z_k z^{-1})}{\prod_{k=0}^{N-1} (1 - p_k z^{-1})}$$

Zeros z_k : roots of $B(z)$, where $|H| = 0$.
Poles p_k : roots of $A(z)$, where $|H| \rightarrow \infty$.
Coinciding pole and zero cancel (reduce order).

Region of Convergence (ROC)

ROC definition

The set of $z \in \mathbb{C}$ for which $X(z)$ converges absolutely. Always circularly symmetric (depends only on $|z|$).

ROC shapes

- FIR** (finite support): entire $\mathbb{C}$, possibly except $z = 0$ or $z = \infty$.
- Causal IIR**: exterior of disk, $|z| > R$, where $R = \max_k |p_k|$.
- Anticausal IIR**: interior of disk, $|z| < r$, where $r = \min_k |p_k|$.
- Two-sided**: annular region $r < |z| < R$. Can be empty.

ROC shape encodes causality. Same $H(z)$ expression with different ROC = different causal nature.

Stability via Poles

BIBO stability $\Leftrightarrow$ unit circle in ROC

LTI is BIBO stable iff unit circle $|z| = 1$ lies inside the ROC of $H(z)$:

$$1 \in \text{ROC}\{H(z)\} \iff \sum_n |h[n]| < \infty$$

For causal systems: BIBO stable $\iff$ **all poles inside the unit circle** ($|p_k| < 1$).

Pole-zero cancellation in cascades

In $H(z) = H_1(z) \cdot H_2(z)$: a zero of H_1 at the same location as a pole of H_2 formally cancels. **Caution:** cancellation is never exact in finite precision — an unstable pole cannot be reliably eliminated this way.

Pole-Zero Plot & Magnitude Response

Pole-zero plot

Plot poles (x) and zeros (•) on the complex plane. Unit circle $|z| = 1$ as reference — frequency response read off along it ($\omega \in [0, \pi] =$ upper half).

Symmetry patterns

Real-valued filter: complex poles/zeros appear in conjugate pairs (p, p^*). Plot symmetric about real axis.

Linear-phase FIR: zeros appear in reciprocal pairs: if z_0 is a zero, so is $1/z_0^*$. Real coefficients: zero z_0 generates quadruple $\{z_0, z_0^*, 1/z_0, 1/z_0^*\}$.

Sketching $|H(\omega)|$ from pole-zero plot

- Zero on unit circle** $\Rightarrow$ exact null at that ω .
- Pole near unit circle** $\Rightarrow |H(\omega)|$ peaks nearby.
- Zero near unit circle** $\Rightarrow |H(\omega)|$ dips nearby.

Running Examples in Z-Domain

Moving Average — z-domain

$$H(z) = \frac{1}{N} \cdot \frac{1 - z^{-N}}{1 - z^{-1}} = \frac{1}{N} \prod_{k=1}^{N-1} (1 - e^{j\frac{2\pi}{N}k} z^{-1})$$

$N-1$ zeros at N -th roots of unity (all except $z = 1$); no poles. ROC: all $\mathbb{C}$ except $z = 0$. Always stable (FIR).

Leaky Integrator — z-domain

$$H(z) = \frac{1 - \lambda}{1 - \lambda z^{-1}}, \quad \text{pole at } z = \lambda$$

ROC: $|z| > \lambda$ (causal). Stable $\Leftrightarrow |\lambda| < 1$ (pole inside unit circle). As $\lambda \rightarrow 1$: pole approaches unit circle, bandwidth narrows.

Filter Realizations

Direct Form I & II

Direct Form I: implement CCDE directly — separate delay lines for x and y ; $M + N$ total delays.

Direct Form II: shared single delay line, $\max(M, N)$ elements. Same transfer function, fewer memory cells.

Second-order sections (SOS / biquads): factor $H(z) = b_0 \prod_k H_k(z)$ into 2nd-order real-coefficient sections, each as Direct Form II. Prevents pole drift in finite-precision arithmetic — *required for high-order IIRs*. (output='sos' in SciPy.)

Intuitive Design (Pole-Zero Placement)

Design principle

Place **zeros on the unit circle** to create exact nulls. Place **poles near the unit circle** to boost gain. Response at ω = product of distances from zeros / product of distances from poles.

DC notch & hum removal

DC removal: $H(z) = 1 - z^{-1}$ — zero at $z = 1$ ($\omega = 0$), but over-attenuates nearby freqs.

DC notch (improved): add compensating pole near $z = 1$:

$$H(z) = \frac{1 - z^{-1}}{1 - rz^{-1}}, \quad r \lesssim 1$$

As $r \rightarrow 1$: notch narrows. Stable iff $r < 1$.

Hum removal: zeros at $e^{\pm j\omega_0}$ with $\omega_0 = 2\pi f_0/f_s$. Add poles nearby at $re^{\pm j\omega_0}$ to control notch width.

Resonator

Simple resonator — poles at $z = re^{\pm j\omega_0}$:

$$H(z) = \frac{1}{1 - 2r \cos \omega_0 \cdot z^{-1} + r^2 z^{-2}}$$

Gain peaks at ω_0 ; bandwidth narrows as $r \rightarrow 1$. Stable iff $r < 1$.

IIR Filter Families

dB scale

$|H(\omega)|_{\text{dB}} = 20 \log_{10} |H(\omega)|$ (reference: passband = 0 dB).
−6 dB: half amplitude; −20 dB: one-tenth amplitude; $A_s = -20 \log_{10} \delta_s$.

Classic IIR lowpass designs

Butterworth: all-pole, monotone (maximally flat) in both bands. Best phase linearity; gradual roll-off. Design: order N , cutoff ω_c at −3 dB.

Chebyshev: equiripple in passband, monotone in stopband. Narrower transition band than Butterworth for same N .

Elliptic: equiripple in *both* bands — most order-efficient (narrowest transition band for given N). Design: $N, \omega_c, \delta_p, \delta_s$.

IIR comparison (same order N)

Narrowest trans. band: Elliptic > Chebyshev > Butterworth
Passband flatness: Butterworth > Chebyshev > Elliptic
Phase linearity: Butterworth > Chebyshev > Elliptic

IIR vs FIR
IIR pros: efficient (low order); strong attenuation IIR cons: stability issues; always nonlinear phase FIR pros: always stable; numerically robust; linear phase FIR cons: high order (long delay); no analog equivalent

FIR Filter Design

Filter specifications
Passband $ \omega \leq \omega_p$: $1 - \delta_p \leq H \leq 1 + \delta_p$ Stopband $ \omega \geq \omega_s$: $ H \leq \delta_s$ Transition band $[\omega_p, \omega_s]$: unconstrained; $\Delta\omega = \omega_s - \omega_p$. $A_s = -20 \log_{10} \delta_s$ (dB).

Windowing method
Truncate ideal (infinite, non-causal) impulse response $h_d[n]$ with a length- M window: $h[n] = h_d\left[n - \frac{M-1}{2}\right] \cdot w[n], \quad n = 0, \dots, M-1$ Shifting by $\frac{M-1}{2}$ makes the filter causal with linear phase (group delay $\frac{M-1}{2}$). In frequency: $H = H_d * W$ — ideal cutoff smeared by window's main lobe. Gibbs phenomenon : rectangular window gives $\approx 9\%$ ripple near cutoff regardless of M . Tapering windows reduce ripple at the cost of wider transition band.

Window functions (length M)		
Window	Atten.	Trans. BW
Rectangular	13 dB	$4\pi/M$
Hanning	44 dB	$8\pi/M$
Hamming	41 dB	$8\pi/M$
Blackman	74 dB	$12\pi/M$
Hanning:	$w[n] = \frac{1}{2}(1 - \cos \frac{2\pi n}{M-1})$	Hamming:
$w[n] = 0.54 - 0.46 \cos \frac{2\pi n}{M-1}$		
Blackman:	$w[n] = 0.42 - 0.5 \cos \frac{2\pi n}{M-1} + 0.08 \cos \frac{4\pi n}{M-1}$	

Windowing design procedure (lowpass)
1. Specify ω_p, ω_s, A_s (dB). Compute $\Delta\omega = \omega_s - \omega_p$. 2. Choose window with attenuation $\geq A_s$; note its BW formula $C\pi/M$. 3. Length: $M \approx C\pi/\Delta\omega$ (round up to odd for Type I). 4. Ideal cutoff at midpoint: $\omega_c = (\omega_p + \omega_s)/2$. 5. Ideal response: $h_d[n] = \frac{\omega_c}{\pi} \text{sinc}(\frac{\omega_c}{\pi} n)$. 6. Apply: $h[n] = h_d[n - \frac{M-1}{2}] \cdot w[n]$. HP: $h_H[n] = (\delta[n] - h_d[n - \frac{M-1}{2}]) \cdot w[n]$. BP: $h_B[n] = 2 \cos(\omega_c n) h_d[n - \frac{M-1}{2}] \cdot w[n]$.

Linear-phase FIR types				
Linear phase requires symmetric (S) or antisymmetric (A) coefficients. Delay $C = \frac{M-1}{2}$.				
Type	M	Sym	Delay	Forced zeros at
I	odd	S	integer	none — most general
II	even	S	half-int	$\omega = \pi$
III	odd	A	integer	$\omega = 0$ and $\omega = \pm\pi$
IV	even	A	half-int	$\omega = 0$
Type I/II: LP and HP ok. Type III: bandpass only (zeros at DC and Nyquist). Type IV: no LP.				

Parks-McClellan (optimal equiripple FIR)
Minimizes the <i>maximum</i> error (minimax, L_∞ norm) in passband and stopband — error is equiripple . Contrast windowing: no control over max error. Inputs: filter length M , ω_p , ω_s , tolerance ratio δ_s/δ_p . Output: M optimal coefficients. Increase M and retry if specs not met. <i>Result:</i> shorter filter than windowing for same specs; works for HP/BP/Hilbert/differentiator.

18 Interpolation & Sampling

The Continuous-Time World

CT vs. DT paradigm

D: $x_d[n] \in \ell_2(\mathbb{Z})$, $n \in \mathbb{Z}$, $\omega \in [-\pi, \pi]$, DTFT
A: $x_c(t) \in L_2(\mathbb{R})$, $t \in \mathbb{R}$ [s], $f \in \mathbb{R}$ [Hz], CTFT

Use cases

- $x(t) \rightarrow$ [processing] $\rightarrow y(t)$
- $x[n] \rightarrow$ [processing for synthesis] $\rightarrow y(t)$
- $x(t) \rightarrow$ [processing for analysis] $\rightarrow y[n]$

Real-world frequency

$f = \text{\#rep/s} = Hz$; $\Omega = 2\pi f$

CTFT (more info in cs)

- NOT PERIODIC
- NO LIMIT FREQUENCY

BandLimited (BL) functions

A CT signal is F_S -BL $\Leftrightarrow X(f)$ for $|f| > F_S/2$

Support of a function

The interval where the function is non-trivially zero

Interpolation $x[n] \rightarrow x(t)$

Interpolation requirement

- decide on T_s (sampling period)
- make sure $x(nT_s) = x[n]$
- make sure $x(t)$ is smooth

19 Quick Reference

Geometric sum

$$\sum_{n=0}^{N-1} r^n = \frac{1 - r^N}{1 - r}, \quad r \neq 1$$

Used in rectangular DFT derivation; also the orthogonality of Fourier vectors (set $r = e^{j\frac{2\pi}{N}(k-h)}$, $k \neq h$).

Euler & trig

- $(e^{j\theta})^* = e^{-j\theta}$
- $\cos(a + b) = \cos a \cos b - \sin a \sin b$
- $\sin(a + b) = \sin a \cos b + \cos a \sin b$
- $\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$
- $\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$
- $\sin a \cos b = \frac{1}{2} [\sin(a + b) + \sin(a - b)]$

Complex number tricks

- $z_1 = z_2^* \Rightarrow |z_1| = |z_2|$ ($|z| = |z^*|$ always)
- $z + z^* = 2 \operatorname{Re}\{z\}$, $z - z^* = 2j \operatorname{Im}\{z\}$
- $z = z^* \iff z \in \mathbb{R} \iff \operatorname{Im}\{z\} = 0$
- $|z_1 z_2| = |z_1| |z_2|$, $\angle(z_1 z_2) = \angle z_1 + \angle z_2$
- $|z|^2 = z \cdot z^*$

Inner product in $\mathbb{C}^N$

Coefficients: $X[k] = \langle \mathbf{w}_k, \mathbf{x} \rangle$ (projection onto basis).

DTFT inner product in $L_2([-\pi, \pi])$

$$\langle \mathbf{F}, \mathbf{G} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} F^*(\omega) G(\omega) d\omega$$

Set $\{e^{-j\omega n}\}_{n \in \mathbb{Z}}$ forms an orthogonal basis.
 $x[n] = \langle e^{j\omega n}, X(\omega) \rangle$

Key warnings

DFT normalization
Analysis: no $1/N$ factor. Synthesis: $1/N$ factor. Convention: $\mathbf{W}^H \mathbf{W} = N \mathbf{I}$.
Phase wrapping
Phase plots show $\angle X[k] \in [-\pi, \pi]$ (wrapped). Apparent discontinuities are $\pm\pi$ jumps, not real.
Real signal DFT
For real $\mathbf{x}$: $X[k] = X^*[N - k]$. Only first $\lfloor N/2 \rfloor + 1$ coefficients are independent. Magnitude symmetric around $k = N/2$.
Circular vs. linear shift
DFT/DFS time shifts are always circular (modulo N). For aperiodic finite signals, multiplying by phase in freq domain gives circular shift.